Score:	/ 10

Name (Print): ______________________

Calculus I (MATH 2200)
Spring 2020
Wednesday, 05/06/2020
Quiz 13: Section 5.3, 5.4

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Section 5.3

2 pts 1. Evaluate the definite integral $\int_0^{10} \left(60x - 6x^2\right)\, dx$ using the Fundamental Theorem of Calculus.

(A) 5000 (B) 1000 (C) 0 (D) 100

2 pts 2. Evaluate the definite integral $\int_0^{2\pi} 3\sin(x)\, dx$ using the Fundamental Theorem of Calculus.

(A) 3 (B) 6π (C) 0 (D) -3

2 pts 3. Use the Fundamental Theorem of Calculus to simplify the expression

$$\frac{d}{dx}\int_1^x \sin^2(t)\ dt.$$

(A) $1+\cos^2(x)$ (B) $\cos^2(x)$ (C) $-\sin(x)\cos(x)$ (D) $\sin^2(x)$

Section 5.4

3 pts

1. A hiking trail has an elevation given by

$$f(x) = 60x^3 - 650x^2 + 1200x + 4500,$$

where f is measured in feet above sea level and x represents horizontal distance along the trail in miles, with $0 \le x \le 5$. Approximately, what is the average elevation of the trail?

(A) 3958.33 ft (B) 11021.67 ft (C) 291.33 ft (D) 11875.00 ft

1 pts 2. Find the point(s) on the interval $[0, 1]$ at which $f(x) = 2x$ equals its average value on $[0, 1]$.

(A) $x = \frac{1}{2}$ (B) $x = 0$ (C) $x = -2$ (D) $x = 1$